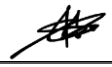
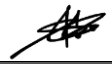
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| | *Approved by:* | Nonhlanhla Msibi -Operations Manager |  |

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**Uncertainty of Measurement**..... 15

## Introduction

Method validation involves determining how well a method performs, understanding its limitations, and identifying the factors that may affect its performance and to what degree. It also ensures that the method is suitable and reliable for addressing a specific analytical task.

## Objective

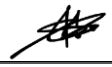
The objective for this document is to validate the proposed procedure AS003WI used to determine the %ash content in coal, ensuring its accuracy, precision, reproducibility, ruggedness, robustness per ISO 1711

## Method Extract

**Ash** determination is a rapid method for estimating the inorganic residue of coal by subjecting a sample to high-temperature combustion (typically 815°C) over a set period (Refer to AS002WI).

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## Samples Matrices

Bituminous Coal- Medium grade coal with carbon content commonly used in power generation.

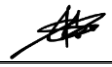
The sample shall be prepared according to instructions on the preparation of sample as per ASOO1IW. The top size for general analysis determination should be < 212 µm.

## Apparatus

- Muffle furnace
- Weighing balance
- Desiccator

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## Performance Characteristics

**Performance characteristics** are the key parameters to be addressed to ensure the method is fit for its intended purpose.

## Accuracy

**Accuracy -** The accuracy of an analytical method is the extent to which test results generated by the method and the true value agree. Accuracy can also be described as the closeness of agreement between the value that is adopted, either as a conventional, true, or accepted reference value, and the value found.

A sample of known value CRM AS002 was analysed 10 times, with the results captured as follows:

| LIMS NO | Original | Duplicate | Average |
|---|---|---|---|
| 1337 | 22.08 | 22.17 | 22.13 |
| 1338 | 22.10 | 21.92 | 22.01 |
| 1339 | 22.59 | 22.54 | 22.57 |
| 1340 | 22.55 | 22.69 | 22.62 |
| 1341 | 22.60 | 22.47 | 22.53 |
| 1342 | 22.36 | 22.74 | 22.55 |
| 1343 | 22.67 | 22.5 | 22.58 |
| 1344 | 22.61 | 22.72 | 22.67 |

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| | | | |
|---|---|---|---|
| 1345 | 22.64 | 22.61 | 22.62 |
| **Overall average** | | | **22.48** |
| **Standard deviation** | | | **0.23559** |

**Percentage recovery**

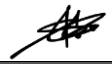
$$\%\text{Percentage Recovery} = \left(\frac{\text{ReferenceValue}}{\text{Validation Average}}\right) * 100$$

$$= \left(\frac{22.50}{22.48}\right) * 100$$

$$= 100.089\%$$

- 100.089% is extremely close to 100%, indicating very high agreement with the reference value.
- This suggest that the method is accurately measuring ash content since it is comparing with the referenced value. This value is well within the commonly accepted accuracy range of 95–105%.

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## Repeatability

**Repeatability** refers to the precision of the method under the same conditions (same operator, equipment, lab, short time frame). It evaluates the consistency of the method when repeated on the same sample**.** It is an acceptable limit between an original and a duplicate of the same sample that is analysed under the same conditions.

| LIMS NO | Original | Duplicate | Average | Difference | 2% Average |
|---|---|---|---|---|---|
| 1337 | 22.08 | 22.17 | 22.13 | 0.09 | 0.443 |
| 1338 | 22.10 | 21.92 | 22.01 | 0.09 | 0.440 |
| 1339 | 22.59 | 22.54 | 22.57 | 0.05 | 0.451 |
| 1340 | 22.55 | 22.69 | 22.62 | 0.14 | 0.452 |
| 1341 | 22.60 | 22.47 | 22.53 | 0.13 | 0.451 |
| 1342 | 22.36 | 22.74 | 22.55 | 0.38 | 0.451 |
| 1343 | 22.67 | 22.5 | 22.58 | 0.17 | 0.452 |
| 1344 | 22.61 | 22.72 | 22.67 | 0.11 | 0.453 |
| 1345 | 22.64 | 22.61 | 22.62 | 0.03 | 0.452 |
| Overall average | | | **22.48** | | **0.450** |
| Standard deviation | **0.23559** | | | | |
| RSD | **1,0482067** | | | | |

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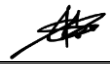
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The repeatability limits and conditions as per ISO 1171 are given as follows:

- Samples that have an ash value that is less than or equal to 10%, the repeatability for such samples is 0.2%
- Samples that have an ash value that is greater than 10% have a repeatability that is given as 2% of the average.

Since the ash values for the reference material used accommodates the second condition, the targeted limit will be 2% of their averages. The averages and their differences are given on the table and all we found within the limits, rendering the method repeatable as per the outlined limits and conditions.

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## Reproducibility

**Reproducibility** is the degree of agreement between results obtained using the same method on identical test material/sample, but under different conditions. These conditions may include different analysts, different laboratories, different equipment or instruments, different days or time intervals. Ash determination has a reproducibility of 3% of the average as its reproducibility.

Reproducibility was addressed by analysis of a sample and comparing the two sets of results analysed by two analysts under the same conditions to check if the method is reproducible.

| Sample no | Analyst 1 | Analyst 2 | Average | Difference | 3% Mean |
|---|---|---|---|---|---|
| 1 | 32,94 | 32,83 | 32,885 | 0,11 | 0,9866 |
| 2 | 32,99 | 32,17 | 32,58 | 0,82 | 0,9774 |
| 3 | 32,78 | 31,92 | 32,35 | 0,86 | 0,9705 |
| 4 | 32,69 | 32,22 | 32,455 | 0,47 | 0,9737 |
| 5 | 32,65 | 32,7 | 32,675 | 0,05 | 0,9803 |
| 6 | 32,59 | 32,77 | 32,68 | 0,18 | 0,9804 |
| 7 | 32,54 | 32,76 | 32,65 | 0,22 | 0,9795 |
| 8 | 32,25 | 31,59 | 31,92 | 0,66 | 0,9576 |
| 9 | 32,16 | 32,89 | 32,525 | 0,73 | 0,9758 |
| 10 | 32,21 | 32,62 | 32,415 | 0,41 | 0,9725 |



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Applying the reproducibility limits as outlined in ISO 1171 set at 3% of the averages, all the results are under their calculated 3% of mean, rendering the method reproducible.

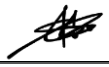
## Precision

Precision- The closeness of agreement between independent test results obtained by replicate measurements in the same or similar samples under specified conditions. It is usually expressed as the standard deviation, relative standard deviation or %RSD.

| | Analyst 1 | Analyst 2 | Average |
|---|---|---|---|
| | 22,13 | 22,24 | 22,19 |
| | 22,01 | 22,04 | 22,03 |
| | 22,67 | 22,83 | 22,75 |
| | 22,62 | 23 | 22,81 |
| | 22,53 | 22,65 | 22,55 |
| | 22,55 | 22,6 | 22,58 |
| | 22,58 | 22,8 | 22,69 |
| **Average** | 22,44 | 22,59 | 22,51 |
| **STD Deviation** | 0,2602 | 0,3412 | 0,2942 |



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**Assigned Value** **22.50**

The precision of the ash determination method was evaluated in accordance with ISO 1171:1997 for bituminous coal. For a certified value of approximately 22.50 % ash, the standard specifies a repeatability limit (r) of ±2 % of the mean, equivalent to ±0.45 % ash. The observed standard deviations from two independent trials were 0.26 % and 0.30 % ash, both within the specified reproducibility limit. These results confirm that the method meets the precision requirements of ISO 1171 and is suitable for the determination of ash content in bituminous coal within the scope of the laboratory's accreditation.

SUMMARY

| *Groups* | *Count* | *Sum* | *Average* | *Variance* |
|---|---|---|---|---|
| Analyst 1 | 7 | 157,09 | 22,44143 | 0,067681 |
| Analyst 2 | 7 | 158,16 | 22,59429 | 0,116395 |

ANOVA

| *Source of Variation* | *SS* | *df* | *MS* | *F* | *P-value* | *F crit* |
|---|---|---|---|---|---|---|
| Between Groups | 0,081779 | 1 | 0,081779 | 0,88853 | 0,364467 | 4,747225 |

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| | | | |
|---|---|---|---|
| Within Groups | 1,104457 | 12 | 0,092038 |
| Total | 1,186236 | 13 | |

**Since F-stat is less than the F-critical, and the p-value is >0.05. There is no significant difference between the two analysts**
**Which implies that the method passes the precision test**
**The method demostrated good precision with the RSD values at 1,16 and 1,51, both of which fall well within general acceptable limits i.e<2 for analytical methods**

## RUGGEDNESS-ROBUSTNESS

The ruggedness study will encompass introducing a slight deviation on the procedure to evaluate how low out of range analysis room temperature affects the accuracy of ash determination. The results will be studied statistically using ANOVA and Welch's T-test.

| LIMS no | Normal Conditions (19-26 oC) | Out Of Range Temperature (15 August 2025) | LIMS no |
|---|---|---|---|
| **1431** | 21,95 | 22,06 | **1476** |
| **1432** | 22,22 | 22,38 | **1477** |
| **1433** | 22,12 | 21,97 | **1478** |



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| | | | |
|---|---|---|---|
| **1434** | 22,05 | 21,94 | **1479** |
| **1435** | 22,03 | 22,03 | **1480** |
| **1436** | 22,08 | 21,80 | **1481** |
| **1437** | 22,06 | 22,19 | **1482** |
| **1438** | 22,19 | 21,83 | **1483** |
| **1439** | 22,20 | 22,08 | **1484** |
| **1440** | 22,22 | 22,09 | **1485** |
| **Mean** | 22,11 | 22,04 | |
| **STD Dev** | 0,0929516 | 0,169708901 | |
| **RSD** | 0,420367223 | 0,770108913 | |
| **ABS Difference** | 0,075 | | |

**Comparison (Normal Conditions VS Stressed Conditions)**

- Δ (Stress – Normal) = **−0.075 ash**
- **Welch t-test**: t = 1.23, p = 0.24 → **not significant** (95 %)
- **95 % CI for Δ**: (−0.206 to +0.056)

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- **ISO 1171 reproducibility limit**: ±0.66 ash
  - |Δ| = 0.075 ≤ 0.66 → **Within tolerance limits**

Anova: Single Factor

SUMMARY

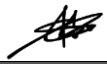
| *Groups* | *Count* | *Sum* | *Average* | *Variance* |
|---|---|---|---|---|
| Normal Conditions (19-26 oC) | 10 | 221,12 | 22,112 | 0,00864 |
| Out Of Range Temperature (15 August 2025) | 10 | 220,37 | 22,037 | 0,028801 |

ANOVA

| *Source of Variation* | *SS* | *df* | *MS* | *F* | *P-value* | *F crit* |
|---|---|---|---|---|---|---|
| Between Groups | 0,028125 | 1 | 0,028125 | 1,502359 | 0,236104 | 4,413873 |
| Within Groups | 0,33697 | 18 | 0,018721 | | | |
| | | | | | | |
| Total | 0,365095 | 19 | | | | |



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Both a **Welch's t-test** (t = 1.23, p = 0.24) and a **single-factor ANOVA** (F = 1.50, F-critical = 4.41, p = 0.236) were performed to compare ash content results obtained under normal ambient conditions (20–25 °C) and stressed conditions (11–15 °C).

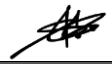
Both statistical approaches showed **no significant difference** between the two groups (p > 0.05), and the observed mean difference (0.075 ash) was also well within the ISO 1171 reproducibility-based acceptance margin (±0.66 3%).

Therefore, the method is demonstrated to be robust against variations in ambient temperature within the tested range.

Welch's t-test was selected as it does not assume equal variances. Calculations were performed in [Excel/Stats software], yielding t = 1.23, df = 16.9, p = 0.24. Since p > 0.05 and |Δ| ≤ 0.45 %, the method is robust to analysis room temperature variation.

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## Uncertainty of Measurement

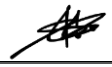
Uncertainty of Measurement is a key element of method validation. It provides an estimate of the doubt associated with a result, considering all potential sources of variability. Evaluating uncertainty ensures that reported values are reliable, comparable and scientifically sound. This section presents the approach used to estimate and express the measurement uncertainty for ash determination.

| | |
|---|---|
| **U of Furnace Calibration(k=2)** | **±4 $^{O}$C** |
| **Operating Temperature** | **±815 $^{O}$C** |
| **U of Furnace Calibration** | **2 $^{O}$C** |
| **UoM** | **0,245398773** |

| | |
|---|---|
| **To calculate Uom of balance** | |
| **UoM on calibration certificate** | **0,00015** |
| **Average mass of sample** | **0,67475** |
| **CRM 002** | **22,5** |
| **UoM** | **0,005001853** |

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2. Uncertainty sources table

| Sources of UoM | Symbol | Standard Uncertainty (U) | U^2 | Divisor | Sensitivity coefficient |
|---|---|---|---|---|---|
| CRM Uncertainty | U1 | 0,135 | 0,018225 | 2 | 1 |
| Balance Calibration | U2 | 0,005001853 | 2,50185E-05 | 2 | 1 |
| Furnace Calibration | U4 | 0,245398773 | 0,060220558 | 2 | 1 |

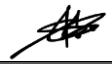
3. Combined Uncertainty

Combined uncertainty (Uc) **0,28**

4. Expanded Uncertainty

Expanded uncertainty (k=2) **0,56**

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5. Final Reporting statement

| | |
|---|---|
| Expanded Uncertainty (k=2) | **0,56** |
| Confidence level: | **95%** |
| Traceability | **All components traceable to SI unit via calibrated instruments** |

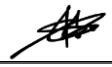
## Conclusion

The validation results for the determination of ash content in coal demonstrate that the method meets the requirements for accuracy, precision, robustness, and reproducibility and UoM as assessed. The observed performance characteristics are consistent with the assigned values of the certified reference material (AS 002) and fall within the acceptable limits specified by ISO 1171:2010 and applicable laboratory quality standards (ISO/IEC 17025).

Accordingly, the method is deemed fit for purpose for routine analysis of ash content in coal, and results generated using this method can be considered reliable and in conformity with international standards.

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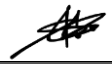
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| **Characteristic (per TR 26)** | **Technical Justification for Omission** |
|---|---|
| **Limit of Detection (LOD)** | **Ash determination is a macro-analytical gravimetric method used to quantify bulk components, not trace levels. The LOD is not a critical parameter because the ash content in the samples (Hardcoal) is significantly higher (typically >1%) than the detection limits of the analytical balance (0.0001g).** |
| **Limit of Quantitation (LOQ)** | **Similar to LOD, the LOQ is not applicable for high-mass gravimetric analysis. The method is designed to measure total inorganic residue where the result is far above the "noise" level of the measurement system.** |
| **Linearity** | **Linearity is applicable to instrumental methods using calibration curves (e.g., ICP or UV-Vis sulphur analyzers). For gravimetric Ash determination, the relationship between the analyte (mass) and the result is inherently linear based on the laws of physics. No calibration curve is generated; therefore, linearity studies are not technically relevant.** |
| **Selectivity / Specificity** | **Selectivity in Ash determination is operationally defined by the method conditions (e.g., incineration at 815°C ± 10°C for coal). The method is non-specific by design, as it measures the total residue remaining after oxidation. Potential interferences are addressed via the temperature control and atmosphere specified in the ISO standard.** |

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| **Characteristic (per TR 26)** | **Technical Justification for Omission** |
|---|---|
| **Stability** | **The analyte (Ash/Inorganic residue) is the stable remains of a combustion process. Once the sample has been cooled in a desiccator to a stable mass, the residue is chemically inert and stable. Stability of the final residue is managed by the "cooling to constant mass" protocol within the procedure itself.** |

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